

Education Session

OTM: Out of the Money

FICC & Quant Club

What we will cover

Ten sections, from the definition of an option to the models used on equity desks

01 **Option Basics**
Calls and puts, European and American, exotics, the straddle

02 **Put-Call Parity**
The no-arbitrage link between calls, puts and the forward

03 **Black-Scholes Model**
The equation, the formula, the Greeks and useful shortcuts

04 **Binomial Model**
One step to N steps, and the limit that gives Black-Scholes

05 **Volatility Revisited**
Realized and implied volatility, forecasting, term structure

06 **Volatility Skews and Surface**
Why implied volatility differs by strike, and how it is quoted

07 **Beyond Constant Volatility**
Local and stochastic volatility, vanna and volga, two underlyings

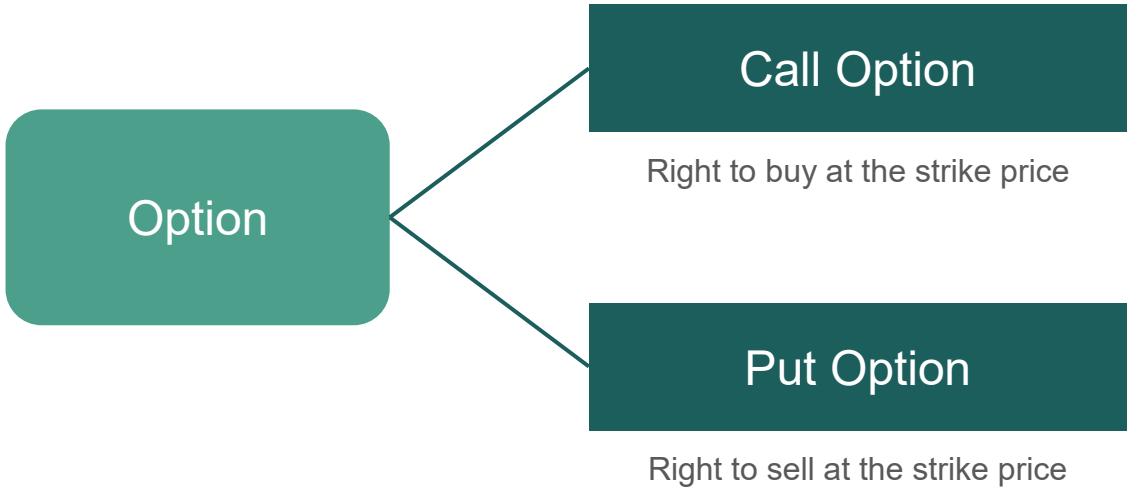
08 **Volatility Contracts**
Variance swaps, the VIX and its derivatives

09 **Equity Models**
Which model for which product, hedging, autocallables

10 **Conclusion**
Options as tools for hedging and speculation

Options: the right, but not the obligation

The holder can buy (call) or sell (put) an asset at a specified price within a set time frame



Analogy: Reserving a concert ticket

You pay a fee to reserve the option to buy a ticket at a s et price. Later, you can buy it at that price, or not at all, d epending on whether you want to attend.

Fee = premium | Set price = strike price

Call: right to buy / Put: right to sell

Both are exercised at the strike price, within the specified time frame

Call Option

The right, but not the obligation, to buy an asset at the strike pric e.

Analogy: Securing the price of a new phone

You pay a fee to lock in a price before the release. If you buy later, you get the secured price, even if market prices have ris en.

Put Option

The right, but not the obligation, to sell an asset at the strike pric e.

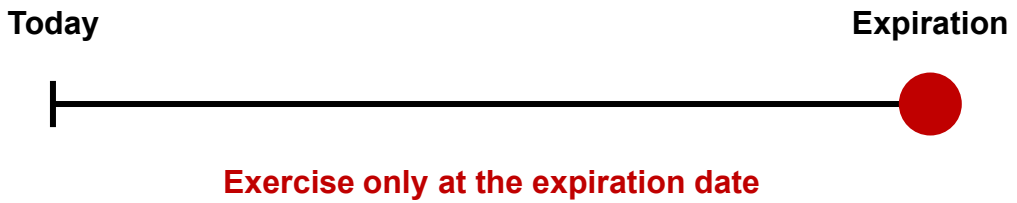
Analogy: A farmer unsure about crop prices

The farmer secures a put option. If prices drop, the crop can still be sold at the decent, pre-agreed price.

European vs. American: when can you exercise?

The two differ only in the timing of exercise

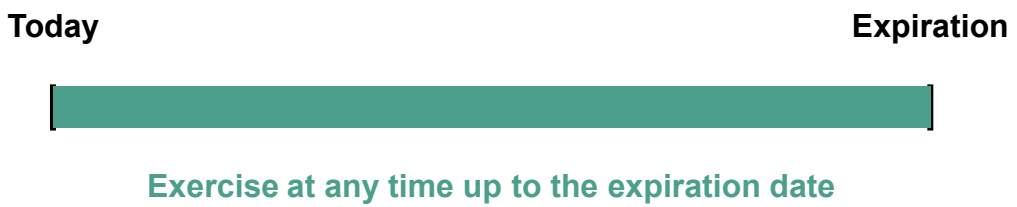
European



Analogy: Restaurant booking token

A booking for an exclusive restaurant that c an only be used on one particular date, not before.

American



Analogy: Amusement park pass

A multi-entry pass you can use to enter the park whenever you want before it expires.

Exotic options: payoffs different from traditional options

Complex options tailored to specific needs, like customizing a car beyond the standard models

Asian

Payoff depends on the average price of t he underlying over a period, not its price at a specific time.

Analogy: Discount coupon

A price reduction based on the averag e price of items bought over a month, not a single purchase.

Barrier

Becomes active or inactive when the und erlying price crosses a predetermined bar rier level.

Analogy: Streaming offer

A special offer that is only available if you have watched a certain number of movies within a month.

Binary (Digital)

Pays a fixed amount if in the money at ex piration, and nothing if out of the money.

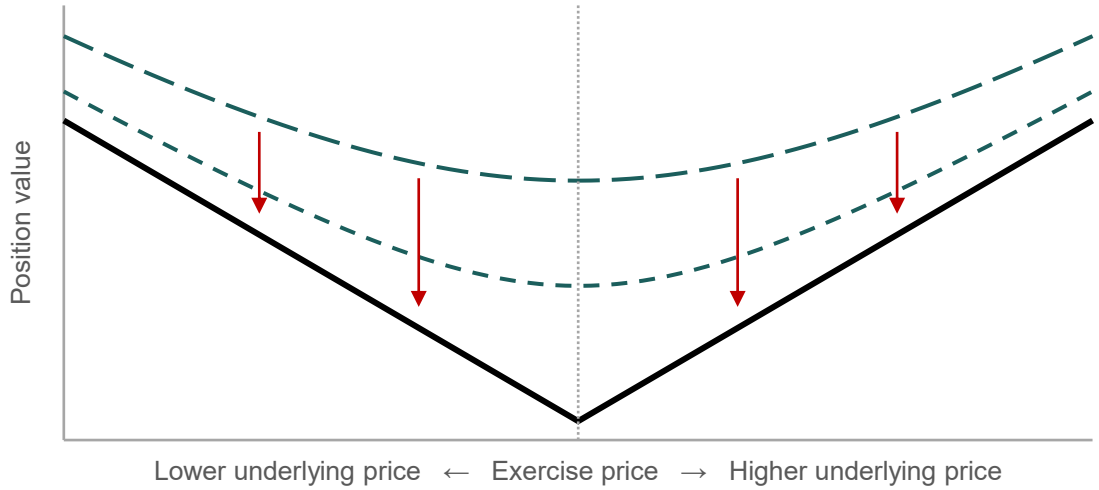
Analogy: Prediction contest

You win a fixed prize if your guess abo ut a future event is correct, and nothin g if it is wrong.

Straddle: a call and a put, same strike and expiration

A position on the size of the move and on volatility, not on its direction

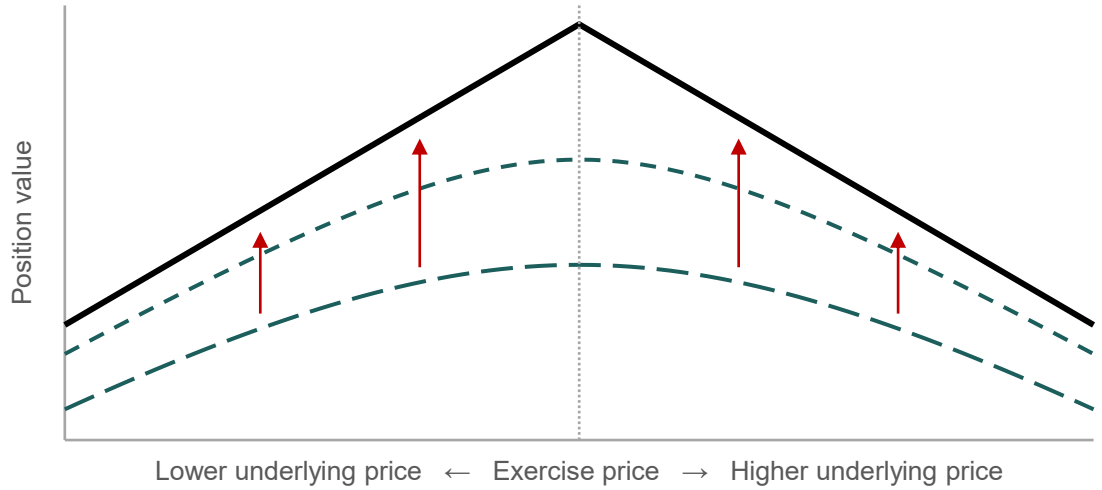
Long Straddle



Buy the call and the put (e.g. +1 March 100 call, +1 March 100 put)

Gains from a large move in either direction. Loses value as time passes or volatility declines.

Short Straddle



Sell the call and the put (e.g. −1 March 100 call, −1 March 100 put)

Gains as time passes or volatility declines. Loses on a large move in either direction.

Solid line: value at expiration. Dashed lines: value before expiration, moving toward the solid line as time passes or volatility declines.

Put-call parity: a no-arbitrage relationship

Same strike and expiration: two portfolios with the same payoff at expiration must cost the same today

Portfolio A
Long call + Short put

= same payoff at expiration =

Portfolio B
Long stock + Borrow PV of X

$$C - P = \frac{F - X}{1 + r \times t} = S - \frac{X}{1 + r \times t}$$

C, P: call and put prices S: stock price F: forward price
X: exercise price r: interest rate t: time to expiration

If C − P is too high: Conversion

Sell the call, buy the put, and buy the stock with borrowed mone y. The position is worth X at expiration whatever the stock does, so the gap is locked in as a riskless profit.

If C − P is too low: Reversal

Buy the call, sell the put, short the stock and lend the proceeds. The mirror-image position locks in the gap in the same way. Arbi trage pushes prices back to parity.

When does put-call parity hold?

Exact equality needs European exercise; the possibility of early exercise turns it into a range

European, no dividend

$$C - P = S - \frac{X}{1 + r \times t}$$

Holds exactly. Any gap is a riskless profit through a conversion or a reversal held to expiration.

European, with dividends

$$C - P = S - \frac{D}{1 + r \times t} - \frac{X}{1 + r \times t}$$

$$\approx S - X + X \times r \times t - D$$

Still holds, once the expected dividends D are taken out of the stock price: the stock holder receives them, the option holder does not.

American

$$S - D - X \leq C - P \leq S - \frac{X}{1 + r \times t}$$

D: present value of dividends (0 if none)

Only a range. A short option can be assigned early, so the position cannot be locked in. Without dividends only the put may be exercised early; with dividends, the call too.

What the approximation tells us

Replacing the discounted values with simple adjustments leaves three intuitive pieces

$$C - P \approx (S - X) + X \times r \times t - D$$

S − X	+ X × r × t	− D
Parity. The difference between buying at X through the options and buying the stoc k at S today.	**Interest on the exercise price.** The call holder pays X only at expiration and keep s the interest until then. Higher rates: calls up, puts down.	**Dividends.** The stock holder receives the m, the option holder does not. Higher divi dends: calls down, puts up.

Approximations used: D/(1 + r×t) ≈ D and X/(1 + r×t) ≈ X − X×r×t. Fine for short maturities and low rates; the error grows with r×t.

In practice: a quick check of conversions and reversals. At the money (S = X), C − P ≈ X×r×t − D. A call and a put at the same strike must carry the same implied volatility.

Black-Scholes: the value of a correctly hedged option

With random stock moves but constant interest rate and volatility, what option price lets a hedged position just break even?

$$rS\frac{\partial C}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + \frac{\partial C}{\partial t} = rC$$

$rS\frac{\partial C}{\partial S}$	+	$\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2}$	+	$\frac{\partial C}{\partial t}$	=	rC
Effect of a change in the stock price (delta)		Effect of curvature and volatility (gamma)		Effect of the passage of time (theta)		Interest on the option value

The equation only states how changes in the stock price S and time t affect the call value C.
To find the exact effect, the equation must be solved: the result is the Black-Scholes formula.

Solving the equation: the Black-Scholes formula

Theoretical value of a European call on a non-dividend-paying stock

$$C = SN(d_1) - Xe^{-rt}N(d_2)$$

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(r + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}$$

$$d_2 = \frac{\ln\left(\frac{S}{X}\right) + \left(r - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}} = d_1 - \sigma\sqrt{t}$$

Inputs	
C	theoretical value of a European call
S	stock price
X	exercise price
t	time to expiration, in years
σ	annual volatility of the stock price
r	annual interest rate
N( )	cumulative normal distribution
ln	natural logarithm

Inside d1: counting standard deviations

Each part of d1 has its own job, and d2 = d1 − σ√t

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(r + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}$$

ln(S/X)	r	σ²/2	σ√t
Moneyness How far the underlying price is from the exercise price: how much the option is in or out of the money	**Carry** Adjusts the spot price to the fo rward price (carry on the unde rlying)	**Lognormal mean** Adjustment for the mean of a l ognormal distribution	**Normalization** One standard deviation: turns the distance into a number of standard deviations

What N(d1) and N(d2) tell us

The call value is an expected value at expiration, discounted back to today

$N(d_2)$	Probability that the option finishes in the money and is exercised
$XN(d_2)$	Average amount we pay at expiration: exercise price × probability of exercise
$Se^{rt}N(d_1)$	Average value of the stock we receive at expiration

$$C = [Se^{rt}N(d_1) - XN(d_2)]e^{-rt} = SN(d_1) - Xe^{-rt}N(d_2)$$

CDF symmetry

$$N(x) = 1 - N(-x)$$

N(−d2): probability that the put finishes in the money

$$P = Xe^{-rt}N(-d_2) - SN(-d_1)$$

One formula, many markets: the cost of carry b

The common variations of the model are determined by the value of b

$$C = Se^{(b-r)t}N(d_1) - Xe^{-rt}N(d_2) \qquad P = Xe^{-rt}N(-d_2) - Se^{(b-r)t}N(-d_1)$$

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(b + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}} \qquad d_2 = d_1 - \sigma\sqrt{t}$$

b = r	**Black-Scholes model:** options on stock
b = r = 0	**Black-Scholes model:** options on futures, futures-type settlement
b = 0	**Black model:** options on futures, stock-type settlement
b = r − r_f	**Garman-Kohlhagen model:** options on foreign currencies (rf: foreign interest rate)
b = r − q	**Dividend-paying stock:** q: annual dividend yield

The Greeks: sensitivities of the option value

Generalized model with cost of carry b; n( ) is the normal density function

		Call	Put
Delta (Δ)		$e^{(b-r)t}N(d_1)$	$e^{(b-r)t}[N(d_1)-1]$
Gamma (Γ)		$\frac{e^{(b-r)t}n(d_1)}{S\sigma\sqrt{t}}$	same for calls and puts
Vega		$Se^{(b-r)t}n(d_1)\sqrt{t}$	same for calls and puts
Theta (Θ)	Call	$-\frac{Se^{(b-r)t}n(d_1)\sigma}{2\sqrt{t}}-(b-r)Se^{(b-r)t}N(d_1)-rXe^{-rt}N(d_2)$	
	Put	$-\frac{Se^{(b-r)t}n(d_1)\sigma}{2\sqrt{t}}+(b-r)Se^{(b-r)t}N(-d_1)+rXe^{-rt}N(-d_2)$	
Rho (ρ)		$tXe^{-rt}N(d_2)$	$-tXe^{-rt}N(-d_2)$

Theta is per year (÷365 for daily decay). Vega and rho are per 100% change (÷100 for one percentage point). Rho shown for b ≠ 0; if b = 0, rho is −tC (call) and −tP (put).

A useful shortcut: the 40% rule

For a European option exactly at the forward, call or put, without a computer

$$\text{Expected value} \approx 0.4 \times F \times \sigma \times \sqrt{t}$$

$$\text{Theoretical value} \approx \frac{0.4 \times F \times \sigma \times \sqrt{t}}{1 + r \times t}$$

Example: X = 65, σ = 18%, 3 months, r = 4%

Expected value: 65 × 18 × √0.25 × 0.00399 ≈ 2.33

Theoretical value: 2.33 / (1 + 0.04/4) ≈ 2.31

The same number for the call and the put: at the forward, put-call parity makes them equal.

Why 0.4?

The peak of the standard normal curve is n(0) ≈ 0.399. One point of volatility is worth about 0.003 99 × X for a one-year at-the-forward option.

One standard deviation

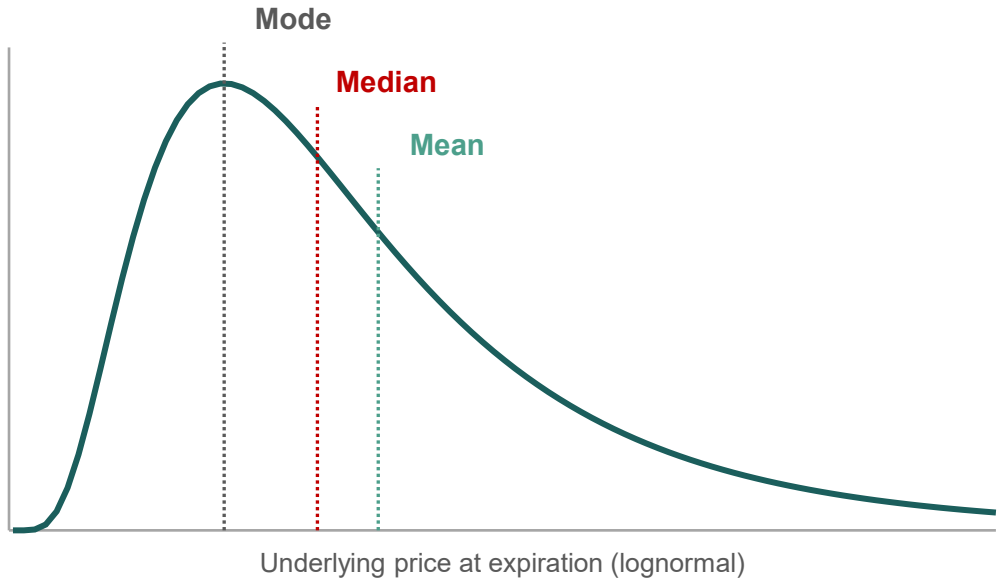
F × σ × √t is a one standard deviation move. The option is worth about 40% of it, and scales with t he strike and with √t.

Limits

Slightly too high for high volatility and long maturi ties, because ATM vega declines a little as volatil ity rises (65 strike, 40%, 2 years: 14.67 vs. 14.48).

Delta is not the probability of exercise

N(d1) is the delta; the probability of finishing in the money is N(d2)



In a lognormal distribution the mean lies to the right of the median. d1 measures from the mean, d2 = d1 − σ√t from the median, so N(d1) is always larger than N(d2).

N(d1) vs. N(d2)	Close for short-dated options, but the delta always overstates the probability of exercise .
At the forward	The call delta is a little above 50, the put a little below 50 in absolute value: an at-the-forward straddle has a positive delta.
Delta-neutral straddle	Needs d1 = 0. The neutral price lies below the exercise price, and further below with more time or volatility.

$$S = Xe^{-\left(r+\frac{\sigma^2}{2}\right)t}$$

Theta, and where the Greeks peak

Theta has three sources; maximum gamma, theta and vega are not exactly at the money

Three sources of theta

Volatility decay	The distribution of prices at expiration narrows. Same for calls and puts, and usually dominant (the "driftless theta").
Spot to forward	For stock, the spot price moves toward the forw ard as time passes: the (b − r) term.
Discounting	The present value of the expected payoff chang es as expiration approaches: the term in r, X an d N(d2).

With zero rates, or futures-type settlement, only the first term remains.

Underlying price where each measure peaks

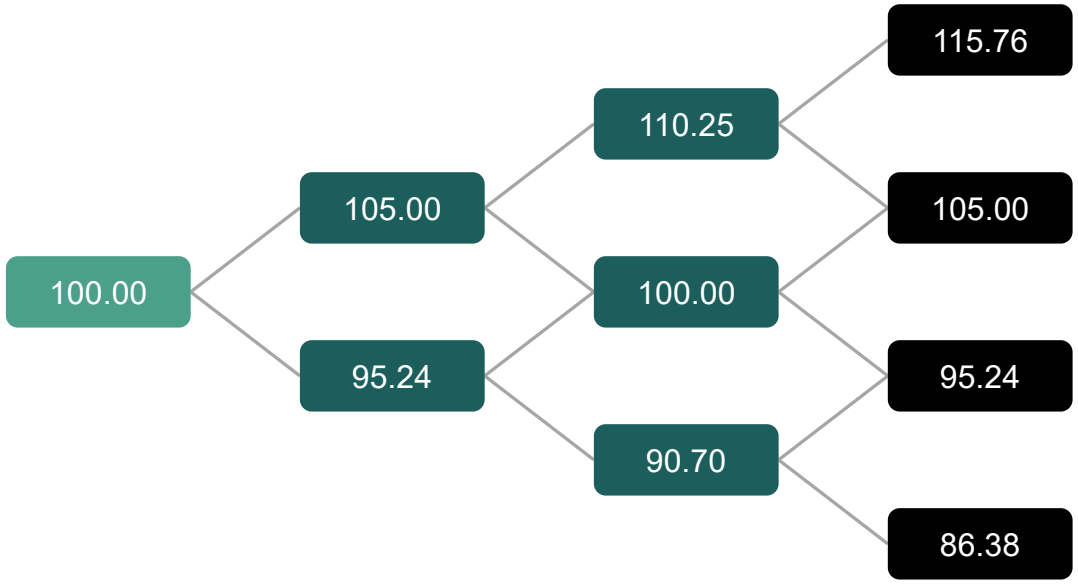
Maximum gamma	$S = Xe^{(-b-3\sigma^2/2)t}$
Maximum theta	$S = Xe^{(b+\sigma^2/2)t}$
Maximum vega	$S = Xe^{(-b+\sigma^2/2)t}$
Delta of 50	$S = Xe^{(-b-\sigma^2/2)t}$

With b = 0: maximum theta and vega sit above the exercise price, maximu m gamma below it.

For stock options with high rates, vega can even fall as time to expiration gr ows (S = X = 100, σ = 20%: beyond 33 months at r = 10%).

Binomial pricing: the same idea in discrete steps

Cox-Ross-Rubinstein: easy to follow, and able to price American options



Example: S = 100, u = 1.05, d = 1/u, three periods over 9 months, r = 4%. p = 0.59; the 100 call is worth 5.22 with a delta of 62.

Up and down

$$u = e^{\sigma\sqrt{t/n}}, \qquad d = \frac{1}{u}$$

One standard deviation per step; recombining

Risk-neutral p

$$p = \frac{1 + r \times t/n - d}{u - d}$$

Makes the expected price equal to the forward

Work backwards

Value at a node = present value of p × up valu e + (1 − p) × down value. Delta = change in op tion value ÷ change in stock price.

American options

At every node take the larger of the hold value and the intrinsic value: early exercise is priced in.

Extending the tree: from one step to N steps

Every path with k up-moves ends at the same node, so the value is a probability-weighted sum of the final payoffs

1 step

$$C = \frac{pC_u + (1-p)C_d}{R}$$

2 steps

$$C = \frac{p^2C_{uu} + 2p(1-p)C_{ud} + (1-p)^2C_{dd}}{R^2}$$

n steps

$$C = \frac{1}{R^n}\sum_{k=0}^{n}\binom{n}{k}p^k(1-p)^{n-k}\max\left(Su^kd^{n-k} - X, 0\right)$$

Check with the three-period tree: only the top two nodes pay (15.76 and 5.00). ($p^3 \times 15.76 + 3p^2(1-p) \times 5.00$) / 1.01^3 = (0.206 × 15.76 + 0.428 × 5.00) / 1.0303 = 5.22.

Growth per step

$$R = 1 + r \times \frac{t}{n}$$

One period of interest at the annual rate r

Paths

There are n-choose-k different paths with k up-moves out of n steps.

Probability

Each such path has the same risk-neutral probability: k up-moves and n − k down-moves.

Payoff

All of them end at the same price, so the payoff depends only on k.

From N steps to the Black-Scholes formula

Cox-Ross-Rubinstein: the sum splits into two probabilities, and as n grows they become N(d1) and N(d2)

$$C = S\Phi(m; n, q) - \frac{X}{R^n}\Phi(m; n, p)$$

$$\Phi(m; n, p) = \sum_{k=m}^{n} \binom{n}{k} p^k (1-p)^{n-k}$$

$$q = \frac{pu}{R}, \qquad 1 - q = \frac{(1-p)d}{R}$$

m is the smallest number of up-moves that finishes in the money (stock price above X). Φ is the pro bability of at least m up-moves. Three-period tree: m = 2, 100 × 0.667 − 97.06 × 0.634 = 5.22.

As n → ∞

$\Phi(m; n, q) \to N(d_1)$

Delta: the share of stock in the hedge

$\Phi(m; n, p) \to N(d_2)$

Risk-neutral probability of exercise

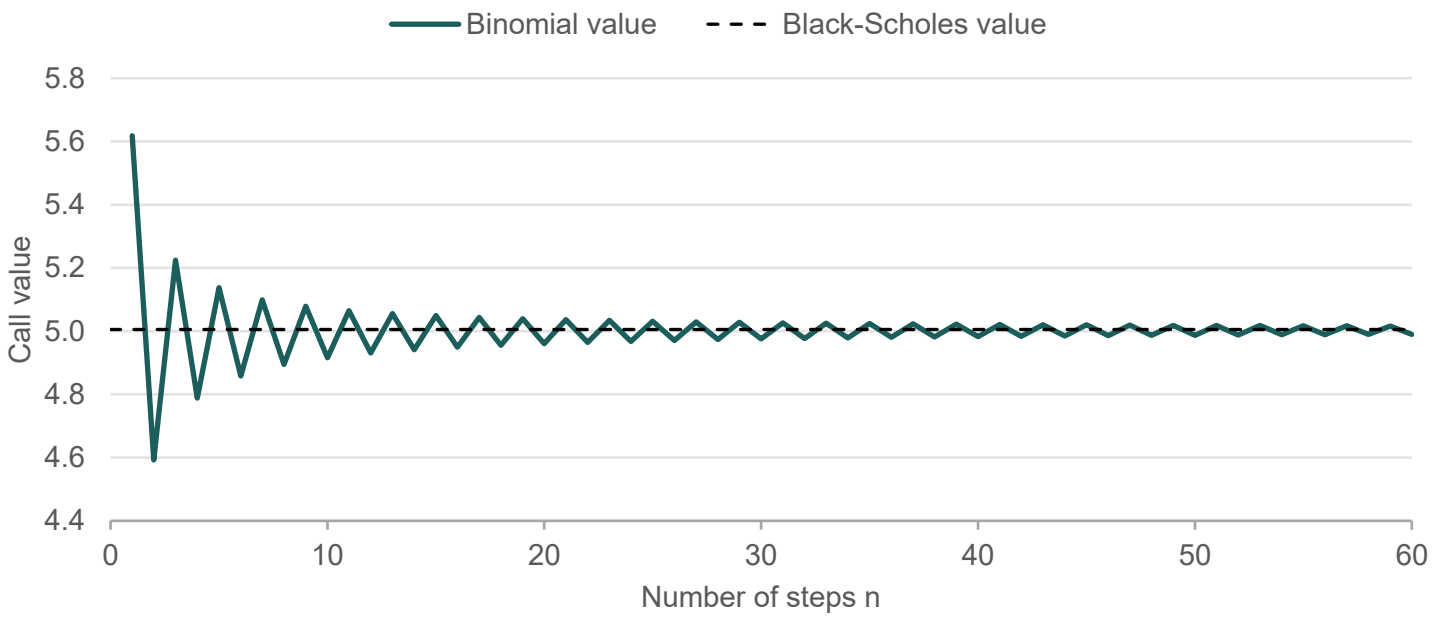
$R^{-n} \to e^{-rt}$

Discount factor

Why normal? The log return to expiration is the sum o f n independent up or down steps. By the central limit theorem it becomes normal, with mean (r − σ²/2)t and variance σ²t.

More steps, same answer: convergence in numbers

The 100 call from the three-period example, repriced with more and more steps



Call value	
n = 3	5.224
n = 10	4.916
n = 100	4.996
n = 1,000	5.004
Black-Scholes	**5.005**

Inputs: S = X = 100, 9 months, r = 4%, volatility 9.76% (implied by u = 1.05 with three periods). The value oscillates because the exercise price falls between two nodes; the swings shrink as n grows.

Realized or implied: which volatility matters?

Example: buy the 100 straddle for 6.25, at an implied volatility of 20%

Implied rises to 22% at once	**Profit 0.62.** The straddle is worth 6.87.
Implied rises to 22% over three weeks	**Loss 0.82.** With the underlying still at 100 the straddle is worth only 5.43: time decay outweighs the rise in implied volatility.
Implied falls to 18% at once	**Loss.** Option prices fall with nothing to offset them.
Implied falls to 18%, underlying moves 5 points	**Profit.** At 95 the straddle is worth 6.87: the positive gamma outweighs the fall in implied volatility.

The longer a position is held, the more realized volatility matters and the less implied volatility matters. Held to expiration, only realized volatility counts.

Implied volatility still decides interim cash flows, capital needs and the result of closing a position early.

Measuring historical volatility

A guide to future realized volatility: interpretation matters more than the estimator

Close-to-close

$$\sigma = \sqrt{\frac{1}{n-1}\Sigma {r_i}^2} \times \sqrt{252}$$

Standard deviation of daily returns. Sample form (n − 1) because it is used as an estimate; the mean is assumed to be zero; annualized with about 252 trading days.

Parkinson: high-low

$$\sigma = \sqrt{\frac{1}{4n\ln 2}\Sigma \ln\left(\frac{h_i}{l_i}\right)^2} \div \sqrt{t}$$

Uses the high and low of each interval, so intraday swings that end unchanged still count. Useful when settlement prices are unreliable.

Garman-Klass: open-high-low-close

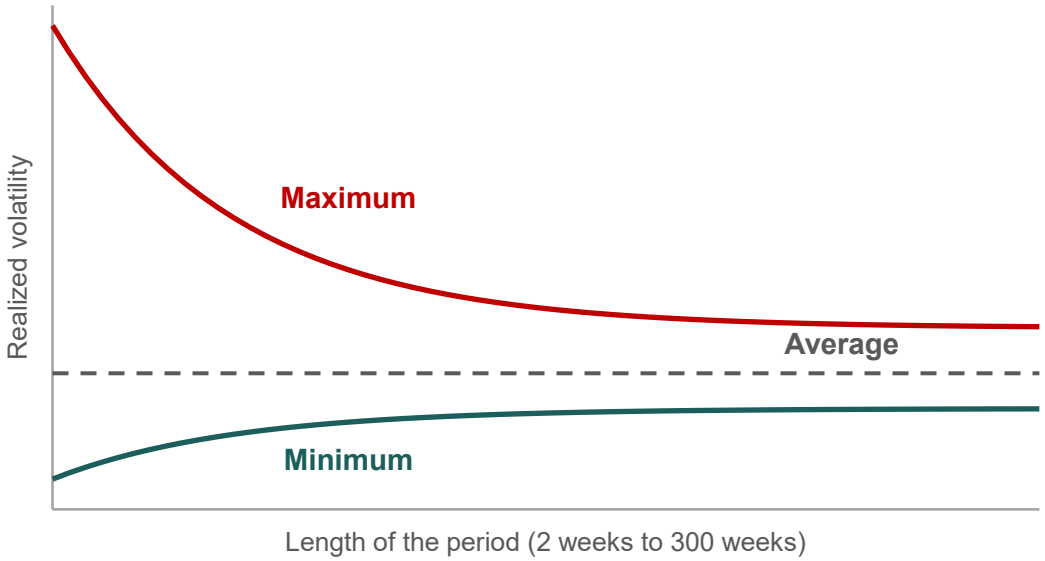
$$\sigma = \sqrt{\frac{1}{n}\Sigma\left[\frac{1}{2}\ln\left(\frac{h_i}{l_i}\right)^2 - (2\ln 2 - 1)\ln\left(\frac{c_i}{o_i}\right)^2\right]} \div \sqrt{t}$$

Adds the open and the close to the high-low range. Like Parkinson it only sees trading hours.

Daily or weekly returns, 252 or 365 days: the results are close. If a contract is volatile from day to day it is volatile from week to week.
For markets open only part of the day, blend the intraday estimate with the close-to-close estimate.

How volatility behaves

Three characteristics that make volatility easier to reason about than price



S&P 500, 2001–2010: two-week volatility ranged from about 5% to over 100 %; 300-week volatility only from about 14% to 24%. The average stayed ne ar 18–19%.

Serial correlation

Like the weather: the next period tends to r esemble the last period of the same length.

Mean reversion

Volatility keeps returning to a long-run level . Prices can trend for years; volatility canno t.

Term structure

Long-term volatility is more stable and easi er to predict. But long-dated options have a larger vega, so a small error costs more.

Forecasting volatility

From weighted historical volatilities to time-series models

History: 6 weeks 28%, 12 weeks 22%, 26 weeks 19%, 52 weeks 18%

Weighting	Weights (6 / 12 / 26 / 52 weeks)	Forecast
Equal	25 / 25 / 25 / 25	21.75%
Recent data first	40 / 30 / 20 / 10	23.4%
5-month option	15 / 25 / 35 / 25: most weight on 26 weeks	20.85%
3-month option	25 / 35 / 25 / 15: most weight on 12 weeks	22.15%

Give the most weight to the history that matches the life of the option. For very long-dated options the best forecast is simply the long-run mean.

EWMA

$$\sigma^2 = \Sigma \alpha_i r_i{}^2$$

Weights on squared returns decay by a factor λ (often about 0.94). Ignores the correlation of returns and mean reversion.

GARCH family

An EWMA-type estimate, plus the tendency of large returns to follow large returns, plus the speed at which volatility reverts to its mean.

Is implied volatility a good forecast?

S&P 500, 2002–2010: implied volatility against the volatility that actually followed

It follows

Implied volatility reacts to what the index has just done. It rose after the index became vol atile in 2008 and fell as the index calmed in 2 009.

Usually too high

In normal conditions implied volatility was ab ove the realized volatility that followed, often by up to 10 points: options look overpriced.

Sometimes far too low

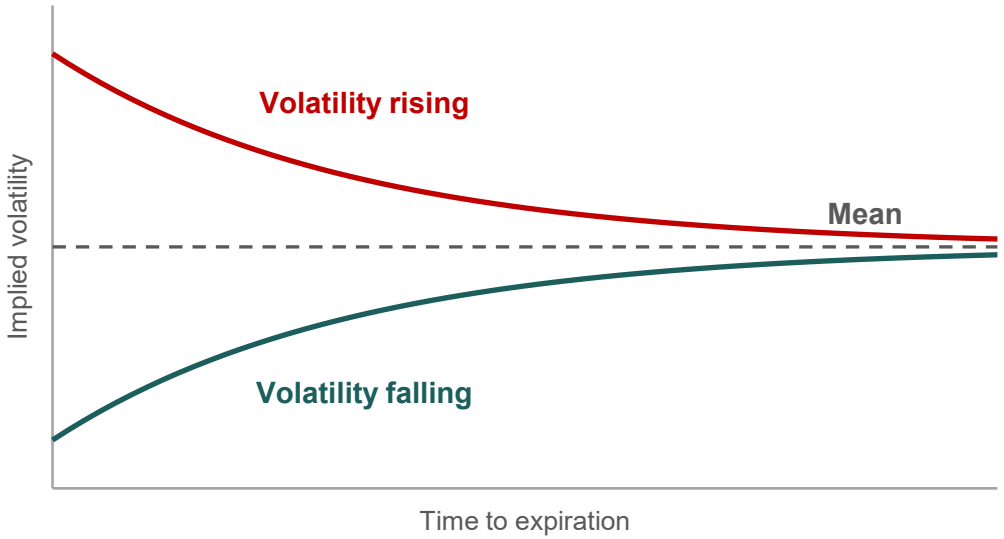
In 2008 the three-month implied volatility wa s at one point almost 50 points below what fo llowed, and at another 20 points too high.

Why do buyers keep paying?

- **Insurance.** A premium is accepted in return for the rare occasions when volatility explodes.
- **Hedging costs.** The seller replicates the option by dynamic hedging and passes the cost on.
- **Model weakness.** Implied volatility comes from a model that does not describe the real world exactly.

Term structure of implied volatility

Short-dated implied volatility moves the most; long-dated stays closer to the mean



A term-structure model needs a primary month, a mean volatility and a "w hippiness" factor. The front month often trades on its own, and seasonal m arkets (natural gas in October, summer crops) break the pattern.

Is a total vega of zero really vega neutral?

Month	Vega	IV change	P&L
April	+15.00	+3.0	+45.00
June	−36.00	+2.0	−72.00
August	−21.00	+1.5	−31.50
October	+42.00	+1.1	+46.20

Total vega 0, P&L −12.30: about −4.1 per point of April volatility

Forward volatility and calendar spreads

Finding expirations that are rich or cheap relative to their neighbours

Forward volatility

$$\sigma_f = \sqrt{\frac{\sigma_2{}^2 t_2 - \sigma_1{}^2 t_1}{t_2 - t_1}}$$

Variance is proportional to time, so two implied volatilities fix the volatil ity the market implies between the two expirations, like a forward inter est rate.

Implied volatility of a calendar spread

$$\sigma_{\text{spread}} \approx \frac{O_2 - O_1}{V_2 - V_1}$$

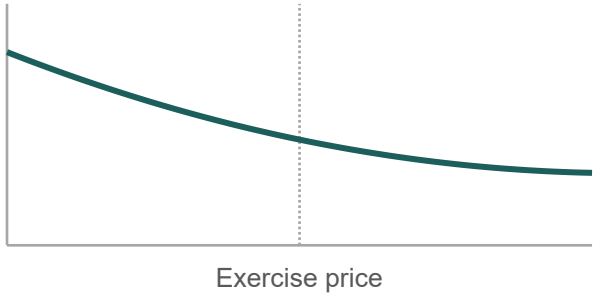
The single volatility that prices both months to the market price of the spread; for at-the-money spreads, roughly the price difference over th e vega difference. February at 29.61% and March at 28.06%: spread volatility 25.94%.

Both act as a magnifying glass for mispricing between months: buy the spread that contains the cheap month and sell the one that contains the r ich month.
Example: buy April/June and sell June/September, together a time butterfly.

Volatility skew: one underlying, many implied volatilities

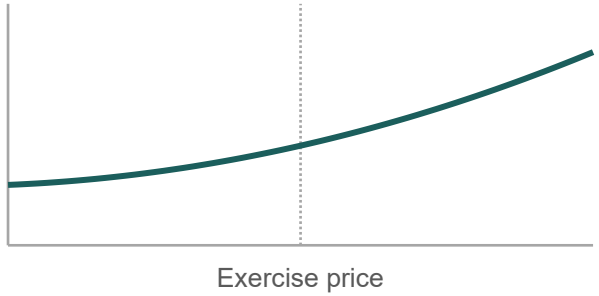
In a Black-Scholes world every exercise price would share one volatility. The market disagrees

Investment skew



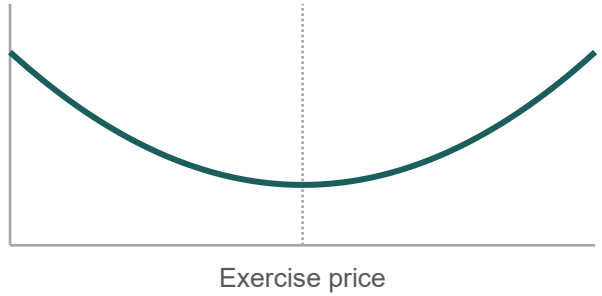
Stocks and indexes. Investors are long: protectiv e puts are bought at low strikes, covered calls ar e sold at high strikes. Low strikes are inflated.

Demand (commodity) skew



Commodities. End users fear rising prices and b uy calls at high strikes. High strikes are inflated.

Balanced skew



Currencies. Both sides hedge, so both wings are inflated, roughly symmetrically: the smile.

Model weakness adds to this: stock markets tend to be more volatile when falling, so a low-strike put gains vega just as volatility rises.
Put-call parity: if the put at a strike is inflated, the call at that strike is too. The skew belongs to the exercise price, not to puts or calls.

Putting the skew into the model

Treat the skew as one more input: a function that gives the volatility at each exercise price

$$y = a + bx + cx^2$$

a — **Base volatility:** the at-the-money implied volatility

b — **Skewness:** the tilt, low strikes against high strikes

c — **Kurtosis:** the curvature, both wings against the money; a lmost always positive

Choosing the axes

x-axis — Not the strike itself but how far it is from the money in stan dard deviations, so that expirations can be compared (a "s ticky-delta" skew).

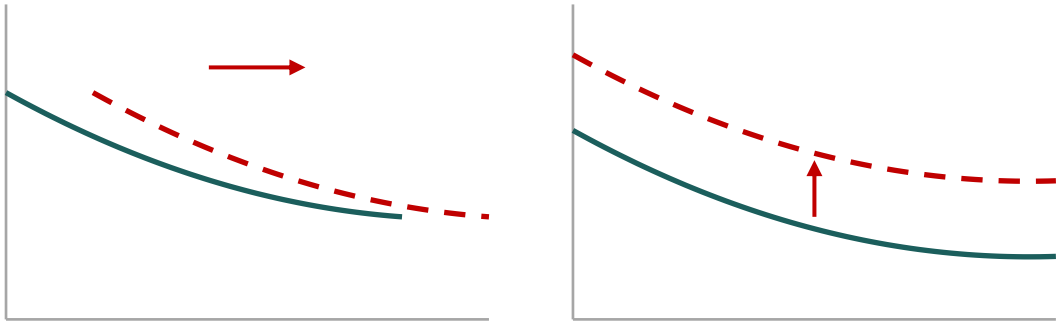
y-axis — Volatility relative to at-the-money volatility: as a difference, or better as a percentage of it.

$$x = \frac{\ln(X/S)}{\sigma\sqrt{t}}$$

Skew risk: the change in value for one unit of b (skewness) or c (kurtosis). The at-the-money strike is the pivot, with zero sensitivity. Skewness risk is largest nea r ±25 delta, kurtosis risk near ±5 delta.

How the smile moves

The skew shifts, and changes shape, as price, volatility and time change



Floating skew: the whole curve moves sideways with the underlying and up or down with at-the-money volatility.

Solid: before. Dashed: after.

Sticky strike	Each exercise price keeps its volatility whatever happens. Simple, but not how markets behave.
Floating skew	The skew travels with the underlying price and with the level of implied volatility.
Passage of time	A fixed strike moves further from the money in standard-deviation terms: an untouched skew looks steeper as expiration approaches.
Change of shape	The tilt (skewness) and the curvature (kurtosis) themselves rise and fall with market conditions.

What the skew does to the Greeks

When the skew moves with the underlying, the option behaves differently from its model Greeks

An out-of-the-money put in an investment skew

Model delta −20	The underlying rises 1.00: the put should lose 0.20.
Skew effect +5	The put is now further out of the money and climbs the skew: +0.5 volatility point × vega 0.10 = +0.05.
Real delta −15	The put loses only 0.15.

Simple approach: take theoretical values from the skew model and the Greeks from the plain model. Large books need skew-adjusted Greeks.

Price and volatility move together

Equity index	Price down, implied volatility up. A "delta-neutral" long straddle is really delta negative: it prefers a falling market.
Commodities	Price up, implied volatility up. The same straddle is really delta positive.
Volatility surface	Skew across exercise prices combined with the term structure across expirations.

Trading the shape of the skew

A view on skewness or kurtosis, separated from direction and from the level of volatility

Skewness: risk reversal

Buy out-of-the-money puts and sell out-of-the-money calls, or the reverse, typically around 25 delta.

Hedge the delta with the underlying. Choose equal vegas so that the position responds to the slope of the skew, not to the level of volatility.

Kurtosis: strangle against straddle

Buy strangles if the wings should become richer; sell them if not.

Offset the vega with at-the-money straddles, which have no kurtosis sensitivity. Two strangles against one straddle is a "dragonfly".

Across expirations

Trade the skew of one month against another: for example buy June OTM puts and sell March OTM puts, with the opposite in calls.

Match the vegas. A view on the two implied volatility levels can be added by choosing which calendar spreads to use.

As conditions change, the delta, gamma and vega of these positions change too: they need to be managed, not just put on.

The distribution implied by option prices

Different volatilities by strike mean the market does not believe in a lognormal distribution

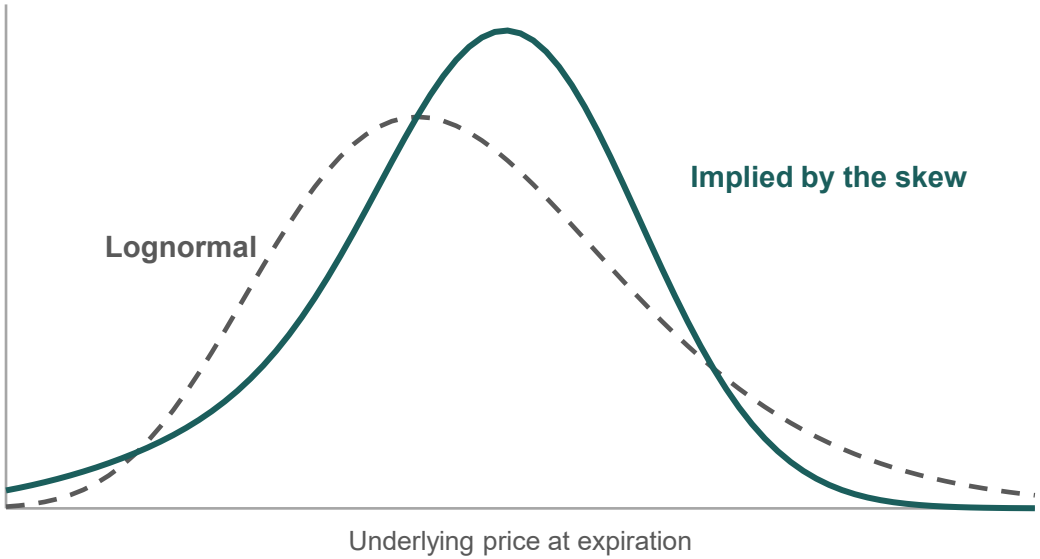


Illustration of a typical stock index. The shape changes whenever option pri ces change.

A butterfly price is a probability

A butterfly pays its full width only if the underlying finishes at its middle strike, a nd all butterflies together always pay exactly the width. So price ÷ width is the p robability of that strike: a 75/80/85 butterfly at 0.15 implies 3%.

Stock index against lognormal

- More small and intermediate upward moves
- Fewer large upward moves
- Fewer small and intermediate downward moves
- More large downward moves

Quoting the surface: ATM, risk reversal, butterfly

Three numbers per expiration describe the level, the tilt and the curvature of the smile

Market quotes

$$\mathrm{RR} = \sigma_{25C} - \sigma_{25P}$$

$$\mathrm{BF} = \frac{\sigma_{25C} + \sigma_{25P}}{2} - \sigma_{\mathrm{ATM}}$$

Back to the wing volatilities

$$\sigma_{25C} = \sigma_{\mathrm{ATM}} + \mathrm{BF} + \frac{\mathrm{RR}}{2}$$

$$\sigma_{25P} = \sigma_{\mathrm{ATM}} + \mathrm{BF} - \frac{\mathrm{RR}}{2}$$

Fitting a curve: SVI

$$w(k) = a + b\left[\rho(k-m) + \sqrt{(k-m)^2 + \sigma^2}\right]$$

w: total implied variance, k = ln(K/F).
a level
b angle between the wings
ρ rotation: the skew, negative for equities
m horizontal shift
σ curvature at the money

No-arbitrage checks

Butterfly: call prices convex in strike (no negative probability).
Calendar: total variance rises with maturity.

Why a constant volatility is not enough

The Black-Scholes-Merton model assumes one fixed volatility for the life of the option

What the model assumes

- Volatility is constant; prices follow a geometric Brownian motion
- Lognormal prices, continuous paths: no jumps
- Continuous trading, no transaction costs or taxes
- A constant, known risk-free rate
- Efficient, arbitrage-free markets; no dividends (easily relaxed)

What the market shows

- **Smile and skew:** implied volatility differs by strike
- **Term structure:** and by maturity
- **Fat tails:** extreme moves are more frequent than normal
- **Clustering:** calm and turbulent periods alternate
- **Leverage effect:** returns and volatility are negatively correlated
- **Regime changes;** long-dated and path-dependent options mispri ced

Still the right tool for: short-dated at-the-money vanillas, calm markets, quick estimates, quoting and benchmarking, basic delta hedging

Three ways to model volatility

Each step relaxes one assumption, and adds complexity

Constant: Black-Scholes-Merton

$$dS = \mu S dt + \sigma S dW$$

Volatility is a single number.

Closed-form prices, fast, the market's quoting language.

But: no smile, no term structure.

Local volatility: Dupire

$$dS = \mu S dt + \sigma(S,t) S dW$$

Volatility is a fixed function of price and time.

Reproduces every vanilla price on today's surface.

But: the surface is static; future smiles come out wrong.

Stochastic volatility: Heston

$$dS = \mu S dt + \sqrt{v} S dW_1$$

$$dv = \kappa(\theta - v)dt + \xi\sqrt{v} dW_2$$

Variance is random, mean-reverting and correlated with price.

Realistic dynamics: clustering, leverage effect, a smile that moves.

But: harder to calibrate; closed forms only for some payoffs.

Local volatility: fitting today's surface

Dupire and Derman: volatility as a deterministic function of the underlying price and time

$$\sigma^2(K,T) = \frac{\frac{\partial C}{\partial T} + rK\frac{\partial C}{\partial K}}{\frac{1}{2}K^2\frac{\partial^2 C}{\partial K^2}}$$

Dupire's formula reads the local volatility directly off observed call pri ces C across strikes K and maturities T.

The pricing equation is the Black-Scholes equation with σ replaced by σ(S, t).

Assumes	A deterministic volatility surface, one source of ran domness, complete markets, continuous paths.
Limits	Static surface and wrong forward smile; calibration is unstable because it differentiates noisy prices; n o mean reversion.
Use for	Barriers, digitals, Asians and other path-dependent exotics; an exact fit to vanillas; risk numbers consi stent with every market price.

Stochastic volatility: letting volatility move

Heston: the price and its variance both follow random processes

$$dS = \mu S dt + \sqrt{v} S dW_1$$

$$dv = \kappa(\theta - v)dt + \xi\sqrt{v} dW_2$$

κ	**Speed** of mean reversion of the variance
θ	**Long-run** variance
ξ	**Volatility of volatility:** fatness of the wings
ρ	**Correlation** of price and variance: the skew

Assumes	Mean-reverting variance correlated with returns (th e leverage effect), continuous paths, constant para meters.
Limits	Complex calibration and unstable parameters; no j umps, so short-dated skew is weak; one volatility f actor.
Use for	Variance and volatility swaps, long-dated options, cliquets and exotics where the dynamics of volatilit y matter.

When volatility moves too: two random factors

Let the volatility follow its own Brownian motion, correlated with the stock price

$$dS = \mu S dt + \sigma S dW_1$$

$$d\sigma = a dt + \xi dW_2$$

$$dW_1 dW_2 = \rho dt$$

ξ: volatility of volatility. ρ: correlation between the stock price and its volatility. a: drift of volatility, for example mean reversion.

Multiplication table

$$(dS)^2 = \sigma^2 S^2 dt$$

Variance of the stock price

$$(d\sigma)^2 = \xi^2 dt$$

Variance of volatility

$$dS d\sigma = \rho \xi \sigma S dt$$

Covariance of the stock price and volatility

One new line for each new source of randomness. All other products are smaller than dt and are dropped.

Itô's lemma in two variables: gamma, volga and vanna

Expanding f(S, σ, t) to second order gives three correction terms instead of one

$$df = f_t dt + f_S dS + f_\sigma d\sigma + \left[\frac{1}{2}\sigma^2 S^2 f_{SS} + \frac{1}{2}\xi^2 f_{\sigma\sigma} + \rho\xi\sigma S f_{S\sigma}\right] dt$$

Gamma	Volga	Vanna
$\frac{1}{2}\sigma^2 S^2 f_{SS}$	$\frac{1}{2}\xi^2 f_{\sigma\sigma}$	$\rho\xi\sigma S f_{S\sigma}$
Curvature in the stock price × variance of the stock price	Curvature in volatility × variance of volatility	Cross curvature × covariance of the stock price and volatility

First-order terms: the coefficient of dS is the delta, the coefficient of dσ is the vega. Every correction is a curvature multiplied by a variance or a covariance.

Hedging two risks, and the equation that results

Two sources of randomness need two hedging instruments: the stock and a second option g

Portfolio	$\Pi = -f + \Delta S + \varphi g$
Delta-neutral	$\Delta = f_S - \varphi g_S$
Vega-neutral	$\varphi = \frac{f_\sigma}{g_\sigma}$
Riskless	$d\Pi = r\Pi dt$

$$f_t + rSf_S + \frac{1}{2}\sigma^2 S^2 f_{SS} + \rho\xi\sigma S f_{S\sigma} + \frac{1}{2}\xi^2 f_{\sigma\sigma} + (a - \lambda\xi)f_\sigma = rf$$

The first three terms are the Black-Scholes equation; the next three are the vanna, volga and vega terms. With ξ = 0 and a = 0 they vanish and Black-Scholes is b ack.

What changes from Black-Scholes

The drift of the stock still disappears, but the price of volatility risk does not

Which drift survives?

Stock: μ drops out	The stock is traded, so its risk is hedged direc tly and its drift becomes r.
Volatility: λ stays	Volatility is not traded. Its drift becomes a − λξ , where λ is the market price of volatility risk.
Calibration	λ is not given by the model. It is fitted to optio n prices, and reflects implied volatility trading above realized.

Which term drives which part of the smile?

Vanna term (ρ)	Risk reversal: the slope of the skew.
Volga term (ξ)	Butterfly: the curvature of the smile.
Vega term (a − λξ)	Term structure of implied volatility.

ρ < 0 (equities): puts become expensive. ρ > 0 (USD/KRW): dollar calls become expensive. A larger ξ makes both wings expensive.

Two assets, constant volatilities: the cross-gamma term

An option on two stocks: two risks, hedged with the two stocks themselves

$$dS_i = \mu_i S_i dt + \sigma_i S_i dW_i, \qquad dW_1 dW_2 = \rho dt$$

$$\Pi = -f + \Delta_1 S_1 + \Delta_2 S_2, \qquad \Delta_i = f_{S_i}$$

Hedge

Two risks, two stocks: no extra option is needed, and both drifts drop out.

Cross-gamma

The new term: cross curvature × covariance of the two stocks.

Correlation

ρ is the input that cannot be observed. Basket, spread, worst-of and quanto options are correlation trades.

$$f_t + rS_1 f_{S_1} + rS_2 f_{S_2} + \frac{1}{2}\sigma_1{}^2 S_1{}^2 f_{S_1 S_1} + \frac{1}{2}\sigma_2{}^2 S_2{}^2 f_{S_2 S_2} + \rho\sigma_1\sigma_2 S_1 S_2 f_{S_1 S_2} = rf$$

Itô's lemma now has three second-order terms: one gamma for each stock, and the cross-gamma multiplied by the covariance of the two stocks. If the payoff dep ends on one stock only, the cross term is zero.

Two assets, stochastic volatilities: four random factors

Each stock now carries its own volatility process; everything can be correlated with everything

$$dS_i = \mu_i S_i dt + \sigma_i S_i dW_i$$

$$d\sigma_i = a_i dt + \xi_i dZ_i, \qquad i = 1, 2$$

Itô's lemma	4 first-order terms and 10 second-order terms : 4 own curvatures and 6 cross terms.
Hedge	Two stocks and two options. Two market pric es of volatility risk remain.

Six correlations, six cross terms

Stock and stock	Cross-gamma: the correlation of the two stocks, as before.
Stock and its own volatility	Vanna of each stock: the skew of each underlyi ng. Two terms.
Stock and the other volatili ty	Cross-vanna: one stock moves, the other's vola tility reacts. Two terms.
Volatility and volatility	Cross-volga: the two volatilities move together.

Two assets, stochastic volatilities: the equation

The same recipe as before: each new curvature is multiplied by its variance or covariance

Two-asset Black-Scholes

$$f_t + \sum_{i=1}^{2} r S_i f_{S_i} + \frac{1}{2}\sum_{i=1}^{2} {\sigma_i}^2 {S_i}^2 f_{S_i S_i} + \rho\sigma_1\sigma_2 S_1 S_2 f_{S_1 S_2}$$

Each volatility on its own: veg a, volga, vanna

$$+ \sum_{i=1}^{2} \left[(a_i - \lambda_i \xi_i) f_{\sigma_i} + \frac{1}{2} {\xi_i}^2 f_{\sigma_i \sigma_i} + \rho_{ii} \xi_i \sigma_i S_i f_{S_i \sigma_i} \right]$$

Cross terms: cross-vanna an d cross-volga

$$+\rho_{12}\xi_2\sigma_1 S_1 f_{S_1\sigma_2} + \rho_{21}\xi_1\sigma_2 S_2 f_{S_2\sigma_1} + \rho_{\sigma}\xi_1\xi_2 f_{\sigma_1\sigma_2} = rf$$

Subscripts on ρ: stock first, volatility second; the first line uses the correlation of the two stocks, the last term that of the two volatilities. Without volatility of volatilit y only the first line remains. With one asset, the first two lines give the single-asset equation.

Why trade volatility directly?

Buying options and delta hedging captures volatility only imperfectly

Problems with dynamic hedging

- The result depends on the order in which price changes occur
- Gaps make it impossible to hedge at the prices the model assumes
- Returns are not normally distributed
- Every rehedge costs a bid-ask spread and fees
- The vega shrinks once the option is no longer at the money

Realized volatility contracts

Settle on the volatility the underlying actually shows over a set period. Mostly traded off-exchange, as variance swaps.

Implied volatility contracts

Settle on the implied volatility of options on a set date. The VIX, with its futures and options, is the main example.

Either way the value at expiration depends only on a volatility calculation: no hedging programme is needed to hold a view on volatility.

Variance swaps: mechanics

Quoted in volatility points with a vega notional, settled in variance points

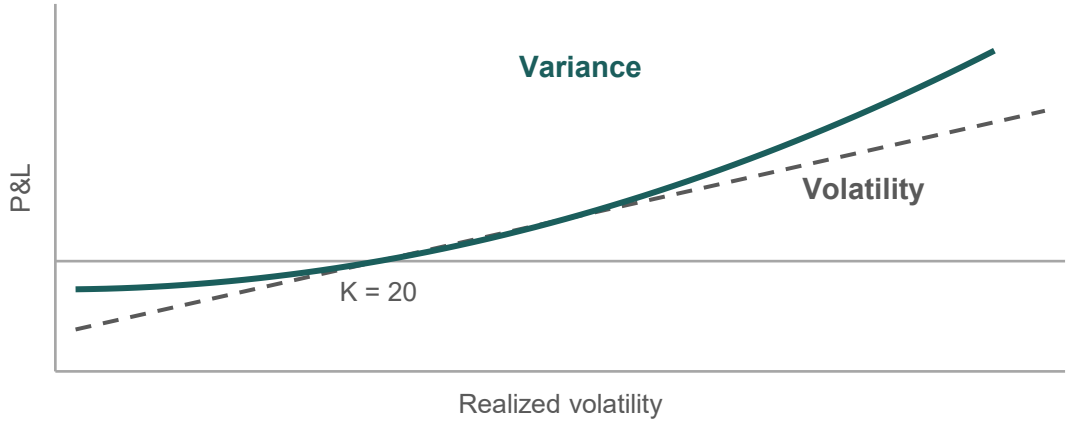
$$\sigma_R = 100 \times \sqrt{\frac{252}{n} \Sigma \ln\left(\frac{p_i}{p_{i-1}}\right)^2}$$

Population form (n, not n − 1) and zero mean: this is the true volatility of the period, not an estimate.

$$\text{Variance notional} = \frac{\text{Vega notional}}{2 \times K}$$

$$\text{P\&L} = \text{Variance notional} \times (\sigma_R{}^2 - K^2)$$

Why variance? It can be replicated with options, and it adds up over time: two months at 25 followed by one month at 22 is √((2 × 625 + 484) / 3) ≈ 24.0 for the three months.

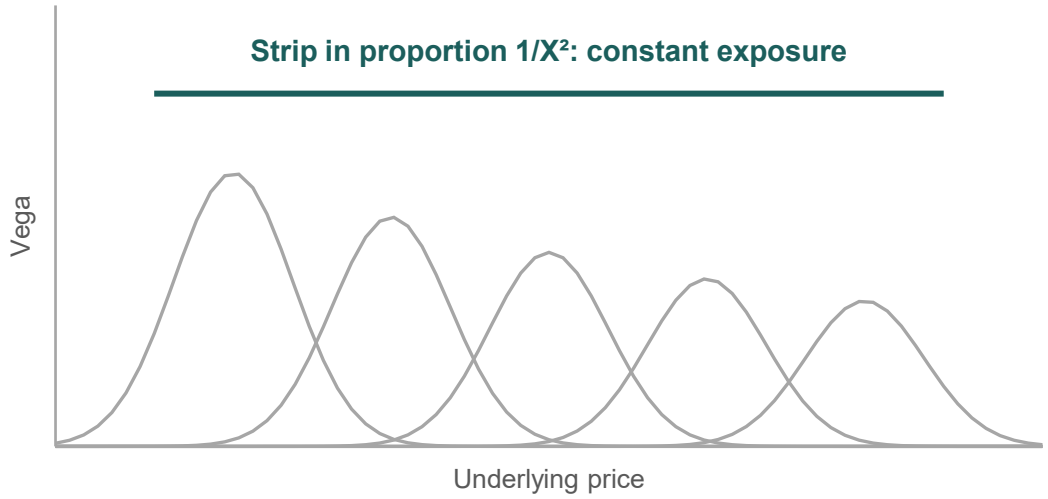


Buy $10,000 vega at 20: $250 per variance point

Realized 19: −$9,750
Realized 23: +$32,250
Realized 50: +$525,000
Capped at 40: at most +$300,000 (caps are common on single stocks)

Replicating variance, and the new VIX

A strip of options weighted by $1/X^2$ has a constant exposure to variance



Grey: single options, whose vega fades away from the strike. More options at low strikes and fewer at high strikes, with the strip delta hedged, reproduce realized variance. The same strip does not give a constant exposure to volatility.

Old VIX, 1993	At-the-money OEX options; implied volatilities from a pricing model, interpolated to 30 days.
New VIX, 2003	The price of a strip of out-of-the-money S&P 500 options across all strikes; two expirations weighted to 30 days.
Model-free	Only option prices and an interest rate are needed. The forward comes from put-call parity.
Settlement	From special opening prices of the options on expiration Wednesday, 30 days before the option expiration.

What the VIX does, and does not, tell us

S&P 500 and VIX, 2003–2012

−0.74

S&P 500 and VIX, daily changes

A strong inverse relationship. The percentage change in the VIX is about 5.7 times that of the index, in the opposite direction.

+0.16

VIX change and the change in realized volatility over the next 30 days

Almost nothing. A rising VIX does not reliably announce a more volatile month.

−0.39

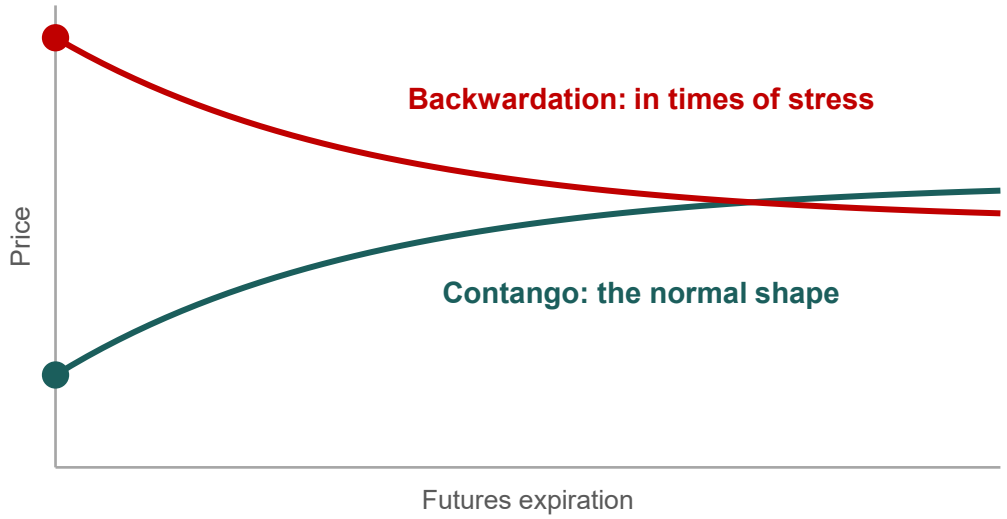
30-day index return and realized volatility

A moderate inverse relationship: falling markets do tend to be more volatile than rising ones.

The VIX is driven less by forecasts of realized volatility than by the demand for protection in a falling market: hence "the fear index".

VIX futures and options

Unlike other futures markets, the underlying index cannot simply be bought or sold



In contango, with no change in the index, a long future slides down the curve and loses value. A spread that buys the far month and sells the near month earns that roll if the curve keeps its shape.

Futures lag the index	July 2011: VIX +4.4, front future +2.0. December 2008: VIX −7.5, front future −5.0.
Why	A future prices where implied volatility is expected to be at its expiration. With no way to hold the index, no arbitrage forces the two together.
Convergence	Only at expiration. The nearer the expiration, the more closely the future follows the index.
VIX options	European, $100 per point, hedged with futures: their implied volatility reflects the calmer future, not the index. Low strikes drop off and high strikes flatten: a "half frown".

Equity market reality: what the models must capture

The forward, dividends, and a skew that is structurally negative

$$F(0,T) = S_0 e^{(r-q)T}$$

Example: index 4,500, r = 4%, q = 1.5%, one year: F ≈ 4,613.9.

A wrong forward is a wrong delta. The forward is the reference point for every Greek.

Dividends are the hidden risk: discrete payments, special dividends and borrow cost all move the forward. On the ex-date the price drops by the dividend.

Unlike rates, equities trend rather than revert, and volatility rises when the price falls.

Why equity skew is negative

Leverage effect	As equity value falls, leverage rises and so does volatility. Spot-volatility correlation is typically −0.6 to −0.9.
Crash risk	1987, 2008, 2020: the market prices gap risk into out-of-the-money puts.
Supply and demand	Investors buy puts for protection; dealers are left short the skew.

Skew is insurance, not arbitrage

A map of equity models

From one constant volatility to hybrids: what each model adds, and what it costs

Model	Core idea	Calibrated to	Strength	Weakness
Black-Scholes, 1973	Constant volatility, lognormal retu rns	ATM volatility	Closed form, fast, the quoting sta ndard	No smile, no skew, no jum ps
Local vol, 1994	Volatility a deterministic function σ(S, t)	The full implied surface	Exact fit to today's surface	Static; wrong forward smile
Heston, 1993	Mean-reverting stochastic varian ce	ATM and skew by maturity	Volatility dynamics; fast through th e characteristic function	Weak short-dated skew; n o jumps
Bates, 1996	Heston plus jumps	OTM puts, crash dynamics	Tails and short-dated skew	Jump parameters hard to p in down
LSV, 2000s	Local volatility scaled by a stocha stic factor	Surface, then exotics	Fit and dynamics together; the str uctured-product standard	Slow (Monte Carlo); unsta ble parameters

Also in use: Variance Gamma (pure jump process), rough volatility (research), and equity-rates-FX hybrids for quantos.

Why Black-Scholes fails but never dies

A quoting convention and a coordinate system, not a belief about the world

Where it breaks

Constant volatility	Volatility spikes in a crisis: losses on short vega .
No jumps	Overnight gaps: the gamma hedge cannot be e xecuted.
Lognormal	OTM puts trade at far higher volatilities: short s kew bleeds.
Continuous hedging	Gaps, slippage and costs leave a residual P&L every day.

Why desks keep it

- Closed form: instant prices
- Every volatility is quoted as a Black-Scholes implied volatility
- A common language for risk reports
- Small error near the money

$$\text{Delta bleed} \approx \text{Vanna} \times \Delta S \times \Delta\sigma$$

The Black-Scholes delta ignores that delta itself changes when volatility moves. Price in Black-Scholes volatility; manage risk with a richer model.

Jumps and local-stochastic volatility

Bates for gaps and short-dated skew; LSV as the desk compromise

Jumps: Merton, Bates

$$dS = (\mu - \lambda k)Sdt + \sigma SdW + (J-1)SdN$$

A Poisson process adds sudden moves: frequency λ, average size an d dispersion of the jump.

Why: earnings, macro shocks and crashes arrive as gaps, and short-dated OTM puts are mostly jump premium.

Hedging: a jump cannot be delta hedged. It is covered by a reserve, not by a trade.

Bates = Heston plus jumps: one model from one week to several year s.

Local-stochastic volatility (LSV)

$$\sigma_{\mathrm{LSV}}(S,t) = \sigma_{\mathrm{loc}}(S,t) \times \sqrt{v_t}$$

Local volatility fits today's surface; the stochastic factor supplies the d ynamics that local volatility lacks.

Calibration in two stages: the local surface to vanillas, then the stoc hastic parameters to exotics.

Cost: heavy Monte Carlo, many parameters, and it is hard to tell whic h part is wrong when the P&L breaks.

Use: the standard for autocallables and other structured products.

Which model for which product?

Use the simplest model that captures the dominant risk

Product	Model	Why
Vanilla, at the money	Black-Scholes / Heston	Vega is the main risk; speed matters; skew is secondary
Barrier, digital	Local volatility	The fit of the surface at the barrier decides the price
Cliquet, forward start	LSV	Depends on the future smile: volatility dynamics are essential
Autocallable	LSV	Knock-out features, skew and path dependence together
Short-dated, earnings, OTM puts	Bates	Jump and tail risk dominate
Variance swap	Heston	Models the variance process directly
Quanto	Equity-rates-FX hybrid	The equity-FX correlation drives the adjustment

Hedging and P&L through the models

Every extra Greek explains part of the P&L that a simpler model leaves unexplained

$$\text{P\&L} \approx \Delta \cdot dS + \frac{1}{2}\Gamma \cdot dS^2 + \text{Vega} \cdot d\sigma + \text{Vanna} \cdot dS \cdot d\sigma + \frac{1}{2}\text{Volga} \cdot d\sigma^2$$

Model	What it hedges	What is left over
Black-Scholes	Delta only	Volatility, skew and gap risk
Local volatility	Delta plus the slide along the surface	Volatility dynamics
Heston	Delta and vega	Jump risk
LSV	Delta, surface vega and factor vega	Parameter instability

A residual that keeps appearing means a missing risk: vanna for delta bleed, volga for the convexity of vega, jumps around earnings. Exotic books hold a model r eserve for this.

Autocallables: where skew, barriers and models meet

The structure behind equity-linked notes such as ELS

Autocallable = Zero-coupon bond + Strip of digital calls − Down-and-in put

The digital calls are the autocall triggers on each observation date; the investor is short the down-and-in put.

Payoff	Redeems early with a coupon if the underlying is at or above the autocall barrier on an observation date. If it is never called and the knock-in barrier has been touched, the investor takes the loss on the underlying.
Source of the coupon	The premium of the down-and-in put the investor has sold: downside risk is exchanged for a coupon well above the risk-free rate.
Hedging the book	Delta, gamma and vega that change quickly near the barriers; correlation for worst-of baskets. Both today's skew and the future smile matter, which is why LSV is used.

Options: flexible tools for hedging and speculation

Their flexibility and diversity let them cater to various needs and strategies



Crucial for effective trading and risk management

Sources and further reading

The worked examples follow Natenberg; the sections on models follow Derman, Miller and Park

Textbooks

Natenberg (2014)

Option Volatility and Pricing: Advanced Trading Strategies and Techniques, 2nd ed. McGraw-Hill Education. Examples, data and figures on put-call parity, Black-Scholes, the binomial model, volatility, skews and volatility contracts (chapters 15, 18–20, 24–25).

Derman, Miller and Park (2016)

The Volatility Smile. Wiley. Variance replication, local volatility, vanna and volga, and the valuation equation with stochastic volatility (chapters 4, 15–17, 19–20).

Original papers

Black and Scholes (1973). The pricing of options and corporate liabilities. Journal of Political Economy 81(3).

Cox, Ross and Rubinstein (1979). Option pricing: a simplified approach. Journal of Financial Economics 7(3).

Garman and Kohlhagen (1983). Foreign currency option values. Journal of International Money and Finance 2(3).

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Hull and White (1987), Journal of Finance 42(2); Heston (1993), Review of Financial Studies 6(2). Stochastic volatility.

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Merton (1976), Journal of Financial Economics 3; Bates (1996), Review of Financial Studies 9(1). Jump-diffusion models.

Margrabe (1978), Journal of Finance 33(1); Stulz (1982), Journal of Financial Economics 10(2). Options on two assets.

Examples taken from these textbooks remain the work of their authors. The two-asset equations with stochastic volatilities are our own extension; any errors are ours.



Thank You